Tail-Value Audits for Rectified-Flow Trajectory World Models

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Abstract

Rectified-flow trajectory generators make it cheap to draw many plausible futures before acting. This paper studies the audit problem created by that convenience: when a learned proxy selects an upper-tail generated trajectory, is the selected future actually better, or has the generator exposed off-manifold artifacts that the proxy overrates? We train a conditional rectified-flow model on a two-mode point-mass world and evaluate selected futures under a deliberately incomplete value proxy with held-out true returns. The audit measures realized return, proxy-realized gap, feature-space distance from the training manifold, selection bias, and mode entropy as candidate budget grows. Across five seeds, proxy-tail selection at the largest budget reduces mean true return from -64.03 for the first sampled candidate to -95.68, while increasing the mean proxy-realized gap to 141.66 and feature-space OOD distance to 321.12. A calibration penalty based on training-trajectory features improves mean true return to -41.57, reduces the gap to 69.14, and reduces OOD distance to 155.68. The claim is intentionally scoped: this is a falsifiable tail audit for rectified-flow world-model trajectories, not a general planning algorithm or evidence about robotics-scale systems.

1 Introduction

Generative world models increasingly make planning look like conditional generation: sample futures, score them, and inspect the most promising trajectory before execution. Rectified-flow models are attractive in this role because a learned vector field can transport base noise into structured continuous trajectories with a small number of Euler steps [1, 2]. The safety question is not only whether average samples look plausible. A planner or analyst often cares about the upper tail selected by a learned proxy, and that tail can be where model errors concentrate.

This paper frames the problem as a *tail-value audit*. Given a conditional rectified-flow trajectory generator and a proxy value model, we ask whether the selected high-proxy futures remain close to the training trajectory manifold and whether their true task return improves with candidate budget. The audit is deliberately modest. It does not claim that rectified flows have a unique pathology, that all generative planning fails at large candidate budgets, or that a feature penalty solves reward hacking. It provides a compact measurement harness for one failure surface: proxy-seeking futures sampled from a flow-generated trajectory pool.

The project is differentiated from the surrounding papers in this collection by its object of study. It is not about language-model answer selection, RSSM latent imagination, memory-augmented rollouts, Bayesian ensembles, differentiable physics, or diffusion policies. The unit under audit is a continuous trajectory generated by integrating a rectified-flow vector field. The main question is whether the chosen upper tail is physically useful or merely far from the data manifold in ways the proxy fails to penalize.

2 Audit Protocol

Let $p_\theta(x \mid c)$ be a conditional rectified-flow trajectory generator for context c. For each context, we sample a candidate pool $\{x_i\}_{i=1}^{N}$ and score each trajectory with a learned proxy $\hat{R}(x_i, c)$. The proxy-tail selector chooses

$$i_{\text{tail}} = \arg\max_i \hat{R}(x_i, c).$$

The calibrated selector subtracts a feature-space distance penalty,

$$i_{\text{cal}} = \arg\max_i \hat{R}(x_i, c) - \lambda u(x_i, c),$$

Table 1: Five-seed audit at candidate budget $N = 16$. Higher true return is better; lower gap and OOD distance are better.

Selector	True return	Proxy-realized gap	OOD distance
First candidate	-64.03 ± 4.23	88.88 ± 6.97	263.79 ± 11.96
Proxy tail	-95.68 ± 5.03	141.66 ± 5.76	321.12 ± 23.84
Calibrated tail	-41.57 ± 1.90	69.14 ± 4.93	155.68 ± 4.95

where u is the standardized distance of trajectory features from the training set and λ is fixed before evaluation. We compare both selectors to the first sampled candidate, which acts as a budget-invariant control.

The audit reports five quantities. True return R is computed by a hidden synthetic evaluator that penalizes endpoint error, obstacle collision, roughness, path length, and backward motion. The proxy-realized gap is $\hat{R} - R$ on the selected trajectory. OOD distance is u. Selection bias is the selected proxy score minus the mean proxy score in the same candidate pool. Mode entropy measures whether selection collapses onto one of the two obstacle-avoidance modes.

Calibration sanity check. If $|R(x_i, c) - \hat{R}(x_i, c)| \leq \beta u(x_i, c)$ for every candidate in a pool, then for any comparison candidate j, the calibrated selector with $\lambda = \beta$ satisfies

$$R(x_{i_{\text{cal}}}, c) \geq R(x_j, c) - \beta u(x_j, c) - \beta u(x_{i_{\text{cal}}}, c).$$

This is a standard robust-selection lens, not the paper's novelty. It clarifies why the penalty must be evaluated empirically: if u is poorly calibrated or misses the relevant proxy error, the inequality gives no useful practical guarantee.

3 Synthetic Rectified-Flow World

The environment is a two-dimensional point mass moving from $(0, 0)$ to a context-dependent goal while avoiding a circular obstacle. Each context specifies goal height, obstacle radius, and lateral wind. Expert trajectories use two modes around the obstacle, which makes diversity meaningful without requiring a large simulator.

The generator is trained as a conditional rectified flow. For data trajectory x_1, Gaussian base sample x_0, and interpolation time t, the vector field predicts

$$v_\theta((1 - t)x_0 + tx_1, t, c) \approx x_1 - x_0.$$

At inference, Euler integration maps Gaussian noise to complete candidate trajectories conditioned on c. The proxy value model is a ridge regressor trained on endpoint and mode-height features only. It intentionally omits direct obstacle-collision, roughness, path-length, and backtracking terms. This makes the proxy accurate enough near the expert manifold but vulnerable to selected upper-tail artifacts.

4 Experiments

We run a smoke-scale experiment for visual diagnostics and a five-seed repeated audit for claims. Each seed trains a conditional rectified-flow generator, fits the proxy on training trajectories, samples evaluation contexts, and scores candidate pools with $N \in \{1, 2, 4, 8, 16\}$. The single-seed figures use seed 0; the table reports mean $\pm$ standard deviation over seeds 0–4 at $N = 16$. All artifacts are generated by the repository scripts.

Result 1: the selected proxy tail is not the best executable tail. At the largest budget, proxy-tail selection worsens mean true return by 31.65 relative to the first candidate and by 54.11 relative to the calibrated selector. The selected proxy score is high, but the true return shows that the selected trajectory is often exploiting omitted features rather than solving the task.

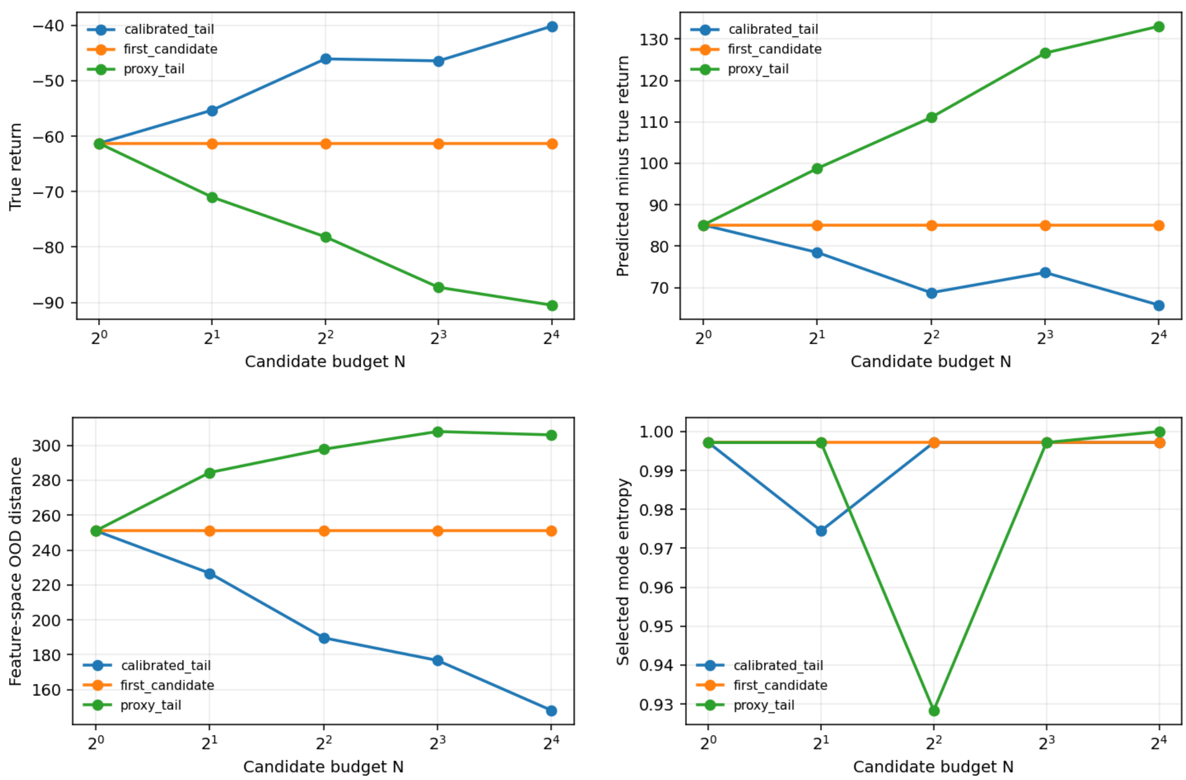

Figure 1: Seed-0 tail audit. Proxy-tail selection increases predicted value but moves selected trajectories toward larger proxy-realized gaps and higher feature-space OOD distance. The calibrated tail reduces those failure signals in this synthetic setting.

Result 2: manifold distance predicts the failure surface. Proxy-tail selection produces the largest feature-space OOD distance and the largest proxy-realized gap. The calibrated selector reduces OOD distance by 165.44 and the proxy-realized gap by 72.51 relative to proxy-tail selection. This supports the audit interpretation: the problem is not only high variance in generated trajectories, but high-proxy selection of trajectories that are far from the feature distribution used to train the proxy.

Result 3: the repair is useful but not a universal solution. The feature penalty improves this synthetic benchmark, but it is not claimed as an optimal planner. It uses hand-built trajectory features, assumes access to a training-manifold statistic, and is evaluated only in a small point-mass world. Its value is to make the audit actionable: when proxy-tail selection fails, the calibrated tail gives a simple falsifiable check that the failure is tied to off-manifold proxy exploitation.

5 Related Work

Flow Matching [1] and Rectified Flow [2] provide the generative modeling foundation for this experiment. Stochastic interpolants and diffusion probability-flow views make clear that rectified-flow samplers share many behaviors with other continuous generative models; this paper therefore avoids claims of flow-specific uniqueness. World-model systems such as PlaNet, Dreamer, TD-MPC, EfficientZero, and DreamerV3 [5] are broader control algorithms than the small audit here. Diffuser-style trajectory generation [4] is a close neighbor because it also treats behavior as conditional trajectory generation. Reward-model overoptimization [3] is the parent phenomenon for proxy-selected tails. The contribution here is the rectified-flow trajectory audit: continuous paths, physical-validity features, candidate-pool diagnostics, and held-out true returns in a reproducible harness.

6 Limitations

The experiment is synthetic and intentionally small. The proxy is deliberately incomplete, not a learned high-capacity value model. The uncertainty score is feature-space distance, not an ensemble, conformal, or flow-consistency uncertainty estimator. There is no standard offline-control benchmark, no robotics or video evidence, and no comparison to diffusion trajectory generators under the same proxy. The right interpretation is therefore narrow: this paper supplies an auditable failure case and a measurement protocol that future work can scale or falsify.

7 Reproducibility

The repository contains the exact scripts used for the results:

- `experiments/run_synthetic.py` regenerates the seed-0 figures.
- `experiments/run_multiseed.py` writes the five-seed aggregate.
- `scripts/run_claim_audit.py` checks the paper's quantitative claims.
- `scripts/build_paper.py` compiles and copies the final PDF.

Unit tests cover selector behavior, metric shapes, diagnostic outputs, and deterministic tiny-run reproducibility.

8 Conclusion

Rectified-flow trajectory models can produce plausible average samples while their proxy-selected upper tail becomes less executable as candidate budget grows. A tail-value audit exposes this by measuring true return, proxy-realized gap, OOD distance, selection bias, and mode entropy on the selected trajectory rather than only on the sample pool. In the controlled setting studied here, a feature-space calibration penalty substantially reduces the failure, but the main contribution is the audit lens and reproducible evidence, not a broad planner.

References

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